

Game Design Document: Fear of Unknown: Invasion

Title: Fear of Unknown: Invasion

Team name: Fear of Unknown Design Group

Team member:

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Permission to Show Your Work in Future Classes:

We hereby grant permission to show this report in future courses as a sample project.

1 Slogan

No matter how you resist, you will be destroyed, eventually.

2 Top-level summary of your game idea and the introduction

Fear of Unknown: Invasion is an Alien-Invasion theme tower defense game. In the previous game, *Fear of Unknown*, the protagonist successfully escapes from the space station to a military base on a planet. In this game, the alien monster continues to evolve and assemble an army to destroy the human race and want to take over their planet. The players will strengthen the defenses of the base and prepare for battle against the invasion. The players must try their best to resist as long as possible.

3 Player experience goal(s) + atmosphere users should experience

Player Experience Goals

- **Survive and Strategize:** Players should feel a deep sense of urgency as they manage limited resources to build and strengthen their defenses. As the game progresses, every decision should have an impact.
- **Creativity:** Encourage players to try different defense strategies and tower combinations. Some innovative thinking will be rewarded. Players can share their experiences in the community.
- **Immersion in the Battle:** Players should feel like a commander, immersed in the desperate defense of humanity's last stand, thinking about strategy and making quick decisions under pressure.

Atmosphere

- **Fear and Tension:** The game should create a palpable sense of unease and tension, with the player feeling desperate as wave after wave of attacks.
- **Futuristic military setting:** The setting should feature an army base on a desolate planet, showcasing humanity's resilience in the face of near extinction.
- **Alien Threat:** Alien enemies should inspire fear and awe, putting players under pressure with unique attacks and abilities.
- **Hope and Sense of Achievement:** Moments of victory—such as successfully surviving after a wave or upgrading a major defense—should give players hope and satisfaction that drives them to continue resisting the invasion.

For the items below, X represents what we want this game to be in scale.

Items from Lazzaro's 4 Keys 2 Fun (2014):

Hard Fun : mastery, achievement, “fiero”:	not at all -----X----- highly so
Easy Fun : Exploring, Imagination, Curiosity, Absorption	not at all -----X----- highly so
Altered States (Serious Fun) : Values & Meaning	not at all ----X----- highly so
People Fun : Relationships, social bonding, etc.	not at all ---X----- highly highly so

Items from Yee's Player Motivation Taxonomy (2006):

Achievement component	
Advancement — The desire to gain power, progress rapidly, accumulate in-game symbols of wealth or status	not at all -----X---- highly so
Mechanics —analyzing the underlying rules and system in order to optimize character performance	not at all -----X----- highly so
Competition — The desire to challenge and compete with others	not at all -----X----- highly highly so
Social component	
Socializing — Having an interest in helping and chatting with other players	not at all -----X----- highly so
Relationship — The desire to form long-term, meaningful relationships with others	not at all -X----- highly so

Teamwork — Deriving satisfaction from being part of a group effort.	not at all X----- highly so
Discovery — Finding and knowing things that most other players don't know about	Immersion component not at all -----X----- highly so
Role-Playing — Creating a persona with a background story and interacting with other players to create an improvised story	not at all --X----- highly so
Customization — Having an interest in customizing the appearance of their character	not at all --X----- highly so
Escapism — Using the online environment to avoid thinking about real-life problems	not at all -----X----- highly so

Summary

The game prioritizes **Hard Fun**, as the game requires experience. Players need to test multiple strategies to last longer during attacks, such as resource management. It also contains **Easy Fun** elements, like exploring the game's background story. **People Fun** (social bonding and relationships) is not included in the current version of the game, as it's a single-player game. However, PVP mode or Cooperation mode may be added in future updates.

From Yee's Player Motivation Taxonomy, the game brings the player a sense of achievement, and the player has the desire to gain power to beat the stronger enemies. Players also want to last longer than others, which is also a competition in a single-player game. Social motivations only appear when a player community is built. The relationships and socializing are less significant, but the players are willing to share their experiences or the strategies they discovered. We hope the game can bring an immersive experience, allowing players to forget their worries in real life and gain happiness in the game.

4 Narrative/Story

After the protagonist escapes the space station, he lands his spaceship on a desolate planet and hides it in a military base. However, The alien from the space station laboratory did not stop hunting. It continued to evolve and assembled an alien army to exterminate humankind. The military base where the protagonist is located has been discovered and is facing an invasion by alien troops. Human beings appear so weak in front of aliens, but they will defend until the last moment.

5 Game mechanics

Resource Management: Players earn resources by defeating enemies or completing waves, which are required to build or repair towers.

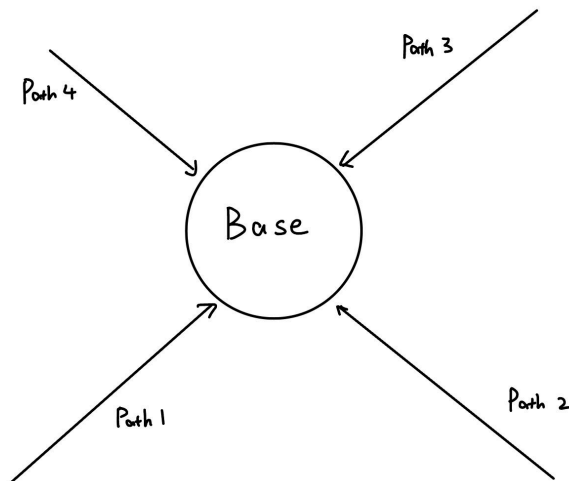
Enemy Pathing: Enemies follow a specific path toward the player's base or target.

Tower Combination: Players can improve towers by putting two or more towers in one spot to have multiple effects.

Tower Placement: Players place towers on four paths to attack enemies.

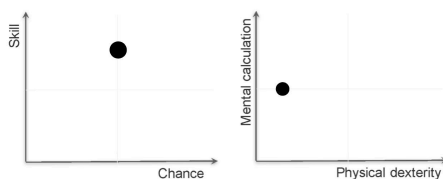
Shock Wave: A large-scale shock wave that causes high damage to all enemies in the field. It has a long charging time.

Enemy Pathing: Enemies get close and attack the base/defensive unit from four directions along the path.



Progressive enemy scaling: Each wave introduces challenges like larger enemy numbers, stronger enemy types with higher health, speed, or special abilities and mixed enemy compositions that require diverse strategies.

6 Play matrix



7 Playtesting script, including testing goals, questions, and assumptions/ hypotheses

Test 1: Atmosphere

Hypothesis:

The game should create a tense and immersive atmosphere, making players feel nervous and requiring strategic thinking to resist for a longer period.

Questions:

1. Did you feel tense or pressured while playing?
2. Could you develop a clear strategy under pressure?
3. Did the visuals, sound effects, or enemy pace contribute to the atmosphere?

Observations:

- Note if players show nervousness (e.g., hesitation, rushed decisions) during gameplay.
- Observe players' reactions during preparation (e.g., building defenses).

Revisions:

- Adjust the enemy spawn pace to enhance tension.
- Refine sound effects and visual cues to improve the atmosphere.

Test 2: Resource Management Mechanics

Hypothesis:

Players should feel a sense of urgency and pressure when managing defensive units effectively under resource constraints.

Questions:

1. Did you feel you had enough defensive units (e.g., rectangle, triangle, circle) to protect your base?
2. Were resource limitations too restrictive, or did they provide a reasonable challenge?

Observations:

- Track how often players run out of resources or units.
- Note whether players express frustration or satisfaction with the level of resource availability.
- Observe if players prioritize specific units over others.
- Observe if players feel the game is too easy/difficult.

Revisions:

- Adjust the cost of the defensive units or the resource generation rate.
- Test the impact of resource constraints to achieve the balance between challenge and enjoyment.

Test 3: Combat Mechanics**Hypothesis:**

Players should critically evaluate their choices and figure out multiple strategies when facing multi-directional enemy attacks.

Questions:

1. Was dealing with enemies attacking from four directions too challenging?
2. Did you feel the defensive units' attack range was sufficient to handle the threats?
3. Did you feel the Tower Combination mechanic is useful?

Observations:

- Monitor if players express difficulty managing enemies from multiple directions.
- Note how players use the Tower Combination mechanic effectively.

Revisions:

- Balance enemy difficulty by adjusting enemies's attributes like speed, health, etc.
- Increase the defensive units' attack range or introduce area-of-effect attacks if players feel overwhelmed.
- Provide clearer guidance or tutorials for using Tower Combination mechanics.

8 Analyze your game in terms of decision types, dilemmas, rewards and endings**Decision Types**

- The player has to decide which unit they want to build; they may need to consider their costs and the unit's function.
- The player must choose where to put the unit to protect the base.

Dilemmas

- Since there are four paths, players may face the problem that if one lane has sufficient defense capabilities, the defense of the other lane may become weak due to insufficient resources.

Rewards

- Players will be rewarded with resources to build defensive units by killing enemies.

Endings

- As the difficulty increases, players will eventually be unable to resist the enemy's attacks and fail. Players will be notified how long they survive.

9 Basic Operation and Sketch of User Interface

Players will use a mouse to control the game throughout the whole game.

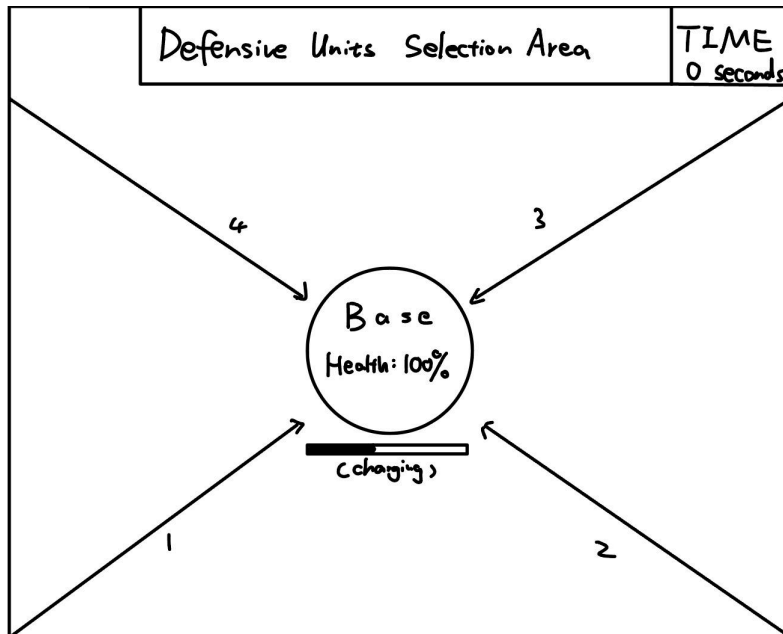
The center of the map is where the player's base is. The player needs to protect the base from enemies' attacks.

The enemy will attack from four paths. Players need to defend their base by building defensive units on these paths.

Players can see the defensive units they can build in the game. It will also show the cost, cooling, and details.

The Energy Bar will be shown at the bottom of the base to remind players how long it takes before they can use a large-scale shock wave to cause great damage to all enemies in the field.

The players can also see how long they have resisted.



10 Tower Defense Game Unit Description

Defensive Unit: Players can place it along the four paths

Unit types: Strength based on its cost. Powerful units require more resources to be built.

Type 1: This type of unit has an effect when it has been placed; it will:

- a. Summon a copy of one unit.
- b. Slow down the enemy for 3 seconds
- c. Poison the enemy for 3 seconds
- d. Fire lasers that can penetrate multiple enemies, dealing damage to all enemies in a line

Type 2: This type of unit has some effect when the enemy destroys it. It has lower health

- a. It can buff Allied units for 10 seconds to make them have higher attack speed and deal greater damage to enemies.
- b. It will explode and instantly kill all enemy targets nearby.
- c. It will control the closest enemy and make it attack other enemies.
- d. It will provide players with extra resources and accelerate the recharge of the base energy bar.

Type 3: This type of unit has some effect when staying on the battlefield.

- a. Deal Area of Effect (AOE) damage to enemies in a particular area for a few seconds.
Then, cool down for a few seconds.
- b. It can heal and upgrade defensive units in a specific area.

Type 4: This type of unit has more health. However, it doesn't do damage. It is for defensive structure. The enemy Unit will attack this unit first.

- a. Have 75 percent more health than a regular unit.
- b. Have 40 percent more health than the regular defensive unit. It will self-heal.

Enemies Unit: Come from four directions

1. This Unit will not cause damage, but it will make a random defensive unit short-out for 5 seconds.
2. This Unit will only target Base
 - a. melee attack
 - b. long-range attack
3. This Unit will target and attack both the defense unit and the Base
 - a. melee attack
 - b. long-range attack
4. This Unit will self-explosion when it touches the first defensive unit. It will do damage in an area and it will move 15 percent faster than regular enemies.
5. Wizard Unit: When the Wizard Unit is alive, it will create a teleport between two areas. The enemies can be teleported across these two areas.
6. Boss Unit: The Boss Unit will evolve by eating other enemy units. It takes a few seconds per evolution.

Phase 1: It will have 90 percent more health than before.

Phase 2: It can do 50 percent more damage than before.

Phase 3: It can have 50 percent more moving speed than before.

Phase 4: It will no longer walk on one path. It can move freely on the map and deal great damage to the defensive units.